

KRASNOUFIMSK, Russian Federation

WMO: 284340

Lat: **56.650N**Lon: **57.783E**Elev: **677**StdP: **14.34**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-25.1	-35.9	0.7	-30.7	-29.8	1.1	-24.9	14.1	19.8	12.1	19.9	0.3	340	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	85.5	67.4	82.2	66.3	78.9	64.4	69.8	82.3	68.0	79.2	66.2	76.1	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	96.3	75.1	63.8	90.9	73.4	62.2	85.9	71.4	34.1	82.5	32.7	79.7	31.3	76.1	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	12.3	10.9	9.5	DB	-39.0	90.0	5.1	3.3	-42.7	92.4	-45.7	94.3	-48.5	96.2	-52.3	98.5
(5)				WB	-39.1	72.0	5.1	2.4	-42.7	73.7	-45.7	75.1	-48.5	76.4	-52.2	78.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.8	4.3	7.1	21.8	38.2	51.9	60.9	64.3	60.2	50.0	37.8	21.5	9.5	(6)
(7)		DBStd	23.84	15.56	14.70	10.07	9.30	8.79	7.54	6.71	7.14	7.60	8.19	13.51	15.51	(7)
(8)		HDD50	6545	1415	1201	874	368	84	9	1	5	92	385	855	1256	(8)
(9)		HDD65	10828	1880	1621	1339	805	413	165	97	186	452	844	1305	1721	(9)
(10)		CDD50	1355	0	0	0	13	144	336	445	320	91	6	0	0	(10)
(11)		CDD65	163	0	0	0	0	8	41	76	36	2	0	0	0	(11)
(12)		CDH74	2033	0	0	0	5	157	539	893	404	35	0	0	0	(12)
(13)		CDH80	530	0	0	0	0	20	138	249	119	4	0	0	0	(13)
(14)	Wind	WSAvg	3.7	3.4	3.8	4.5	4.5	4.3	3.3	2.7	2.6	3.1	4.3	4.1	3.7	(14)
(15)	Precipitation	PrecAvg	20.9	1.3	0.9	0.7	1.2	1.6	2.6	3.0	2.5	2.2	1.9	1.7	1.3	(15)
(16)		PrecMax	32.0	3.0	3.3	2.3	3.3	3.1	5.1	7.1	4.9	4.8	4.6	3.5	2.6	(16)
(17)		PrecMin	12.2	0.1	0.0	0.0	0.1	0.1	0.7	0.7	0.9	0.4	0.4	0.4	0.1	(17)
(18)		PrecStd	3.7	0.6	0.7	0.5	0.8	0.7	1.1	1.7	1.0	1.1	0.9	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	35.8	49.1	72.7	82.5	88.4	89.4	88.9	79.5	65.0	47.6	35.3	(19)
(20)			MCWB	32.7	33.0	40.2	54.0	60.1	67.5	69.9	69.0	60.6	52.6	43.6	33.5	(20)
(21)		2%	DB	31.8	33.3	42.0	65.8	78.5	83.7	85.8	83.1	72.6	59.0	42.5	33.7	(21)
(22)			MCWB	30.2	31.2	36.0	50.7	59.4	66.4	68.4	67.6	58.9	49.3	40.0	32.1	(22)
(23)		5%	DB	29.2	30.9	38.2	59.3	74.4	80.2	82.8	78.2	67.3	54.3	38.8	32.0	(23)
(24)			MCWB	27.7	28.9	33.9	47.5	57.1	64.7	67.6	65.4	57.0	47.8	37.0	30.6	(24)
(25)		10%	DB	25.9	28.0	35.5	53.4	70.0	76.3	79.3	73.6	62.7	49.9	35.8	29.2	(25)
(26)			MCWB	24.4	26.0	32.4	44.1	55.6	63.4	66.1	63.6	55.4	45.7	34.2	27.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	33.7	40.6	55.5	64.4	70.8	72.1	71.1	63.9	54.2	44.1	34.1	(27)
(28)			MDCB	34.7	35.1	47.8	72.0	77.3	83.1	84.0	83.8	73.3	60.9	46.2	34.8	(28)
(29)		2%	WB	30.2	31.5	36.7	51.4	60.9	68.2	70.1	68.8	60.3	50.8	40.4	32.4	(29)
(30)			MDCB	31.7	33.2	40.4	63.7	74.2	79.8	82.6	80.2	68.3	57.7	41.9	33.5	(30)
(31)		5%	WB	27.8	29.1	34.7	48.3	58.9	66.3	68.7	66.4	58.0	48.3	37.1	30.4	(31)
(32)			MDCB	29.2	30.8	37.6	58.4	71.3	77.0	80.5	76.2	65.6	52.4	38.6	31.8	(32)
(33)		10%	WB	24.4	26.0	32.6	45.1	56.7	64.3	67.0	64.2	55.9	45.9	34.5	27.8	(33)
(34)			MDCB	25.8	27.9	35.3	52.3	67.7	74.3	77.3	72.0	62.0	50.2	35.6	29.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.0	19.1	19.1	19.2	22.3	20.7	18.7	16.8	11.7	10.7	13.2	(35)	
(36)			MCDBR	11.0	13.2	18.4	29.4	31.1	28.5	28.1	27.1	25.1	23.6	10.1	9.4	(36)
(37)			MCWBR	10.4	11.5	13.4	16.5	15.0	13.0	12.6	12.9	13.3	14.8	8.3	9.3	(37)
(38)		5% WB	MCDBR	10.6	11.8	15.3	27.8	26.3	25.2	25.1	25.5	21.4	20.2	9.2	9.7	(38)
(39)			MCWBR	10.2	10.7	11.8	16.0	14.1	12.4	12.0	13.0	12.1	13.7	8.1	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.318	0.374	0.409	0.403	0.397	0.420	0.438	0.383	0.353	0.313	0.290	(40)	
(41)		taud	2.013	2.017	1.927	2.046	2.247	2.328	2.264	2.185	2.308	2.254	2.108	1.979	(41)	
(42)		Ebn at Noon	173	222	238	252	265	268	258	242	240	212	166	136	(42)	
(43)		Edh at Noon	20	31	44	45	40	37	39	40	30	24	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	205	502	925	1318	1620	1725	1726	1313	807	355	167	119	(44)	
(45)		RadStd	28	70	94	164	181	172	183	137	70	71	33	25	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)